Math 526, F06 **Name:**
Quiz 3 **SSN: xxx-xx-**

<u>Instructions:</u> **There are totally two problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (6 points) Find a 3×3 matrix $\mathbf{A}$ so that, for any $3 \times n$ matrix $\mathbf{B}$ with rows $\mathbf{b}_1$,$\mathbf{b}_2$ and $\mathbf{b}_3$, (i) the first row of $\mathbf{AB}$ is $\mathbf{b}_1 - 3\mathbf{b}_3$, (ii) the second row of $\mathbf{AB}$ is $-\mathbf{b}_2 + \mathbf{b}_3$, (iii) the third row of $\mathbf{AB}$ is $-4\mathbf{b}_1 + 5\mathbf{b}_2 + 3\mathbf{b}_3$.

Solution:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -3 \\ 0 & -1 & 1 \\ -4 & 5 & 3 \end{bmatrix}$$

Problem 2. (14 points) Let $\mathbf{A} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 1 \\ -2 & 4 & -2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -1 \\ 0 \\ 10 \end{bmatrix}$.

a) Find the $\mathbf{L} - \mathbf{U} - \mathbf{P}$ factorization of $\mathbf{A}$.

Solution:

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 1 \\ -2 & 4 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ -2 & 4 & -2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & & \\ 0 & & \\ -2 & & \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & & \\ 0 & & \\ -2 & 2 & \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So we have

$$\mathbf{L} = \begin{bmatrix} 1 & & \\ 0 & 1 & \\ -2 & 2 & 1 \end{bmatrix}, \qquad \mathbf{U} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}, \qquad \mathbf{P} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

b) Using the $\mathbf{L}-\mathbf{U}-\mathbf{P}$ factorization from (a) to solve $\mathbf{Ax}=\mathbf{b}$ where $\mathbf{x}=\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

Solution:

First step: Solve $\mathbf{Ly}=\mathbf{Pb}$. It is easy to find

$$\mathbf{Pb}=\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} -1 \\ 0 \\ 10 \end{bmatrix}=\begin{bmatrix} 0 \\ -1 \\ 10 \end{bmatrix}$$

So we solve

$$\begin{bmatrix} 1 & & \\ 0 & 1 & \\ -2 & 2 & 1 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}=\begin{bmatrix} 0 \\ -1 \\ 10 \end{bmatrix}$$

We get

$$y_1=0, \quad y_2=-1, \quad y_3=12.$$

Second step: Solve $\mathbf{Ux}=\mathbf{y}$.

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} 0 \\ -1 \\ 12 \end{bmatrix}$$

We have

$$x_3=6, \quad x_2=5, \quad x_1=-1.$$

So we have

$$\mathbf{x}=\begin{bmatrix} -1 \\ 5 \\ 6 \end{bmatrix}.$$